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Title / Título

FALCON 10X HTP CERTIFICATION PLAN

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Summary / Resumen

This document is the structure certification plan of the Falcon 10X WP55-A Horizontal Stabilizer. This package comprises all structural parts and system brackets that constitute the Horizontal Stabilizer. This plan describes the analyses and tests planned to demonstrate the conformity of the mentioned work package with the requirements of EASA/FAA airworthiness regulations.

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1 Object

This document is the structure certification plan of the DASSAULT Falcon 10X WP55-A Horizontal Stabilizer. This work package, described in Ref. 1, comprises all structural parts and system brackets that constitute the Horizontal Stabilizer.

This plan describes the analysis and tests planned to demonstrate the conformity of the Horizontal Stabilizer structure to the requirements of the EASA/FAA airworthiness regulations (Ref. 4) and to the Future Falcon Strength Substantiation Directive (Ref. 2).

1.1 References

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	DGT 164919 - APPENDIX B2	B	High Level Technical Requirement for Future Falcon Horizontal Stabilizer.
Ref. 2	DGT 161868 - APPENDIX B13	A	Future Falcon Strength Substantiation Directive.
Ref. 3	DGT 167943 - APPENDIX B1-2	B	Future Falcon Airframe General Technical Requirements.
Ref. 4	DGT164380 - APPENDIX B15	2	Certification Requirements.
Ref. 5	TEAS-NT-1910014	F	Material Design Values and Methods for Structural Analysis of the Future Falcon.
Ref. 6	[EASA] AMC 20-29		Composite Aircraft Structure.
Ref. 7	F46 EP061	A	PSE Selection Criteria.
Ref. 8	TEAS-NT-1810012	D	Falcon 10X VTP Certification Plan.
Ref. 9	TEAS-PR-2110008	B	Certification of composite structures – Airbus D.S. experience in Category 3 damages justification.
Ref. 10	ECM F46.DA.01757/21		F46- Safety of flight process - Guidance for airframe product packages.
Ref. 11	TEA-F9-NT-210021 (Draft)	A	Falcon 10X Empennage – Lightning Direct Effects Certification Test Plan

1.2 Structure Description

The Falcon 10X Horizontal Stabilizer is an all-moving control surface type (adjustable angle of attack) including a main structural assembly and two pitch control surfaces called elevators.

The main components of the WP#55-A Horizontal Stabilizer are:

- The horizontal stabilizer box structure that is the main structure (skin-stringer cocured configuration, spars and ribs). The stabilizer is hinged around rear attachment lugs and actuated by an actuator linked to front attachment lugs. A mechanical device ensures the lateral guidance of the stabilizer.
- Four removable leading edges, two on each side.
- Two removable tip fairings: one on each side.

- Two spring loaded fairings mechanism with the leading edge extensions streamlining the junction between the horizontal and the vertical stabilizer. The fairings and leading edge extensions both slide onto the fin.
- Two independent elevators, each one on each side, and actuated by a dual hydraulic servo-actuator (located inside the horizontal stabilizer box structure).

The geometrical definition of the Falcon 10X Horizontal Stabilizer is shown in next figure:

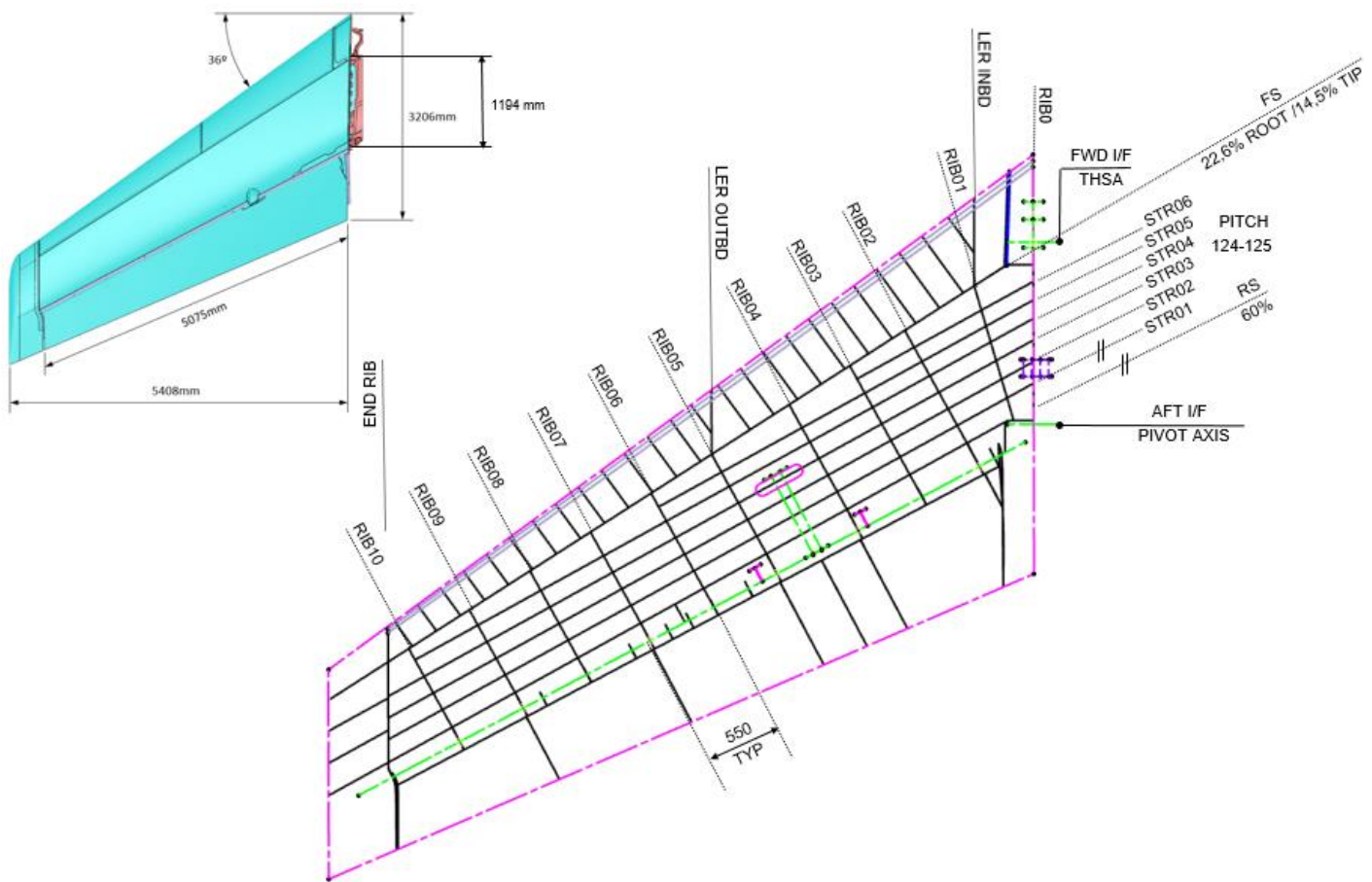


Figure 1. Geometrical definition of the Falcon 10X Horizontal Stabilizer

1.3 Boundaries of the Structure

The Falcon 10X Horizontal Stabilizer has interfaces with others airframe and aircraft systems Product Packages. Each interface shall be managed through Interface Control Documents.

The main Horizontal Stabilizer structural interfaces are:

- Vertical fin and Rudder
- Flight control system

2 Strength Justification Plan

The compliance of the Falcon 10X Horizontal Stabilizer Structure with the applicable requirements will be demonstrated by means of analysis supported by test evidences and experience on similar structures. This plan is based on DASSAULT Future Falcon Strength Substantiation Directive (Ref. 2) and Airframe General Technical Requirements (Ref. 3).

Analysis will demonstrate that all applicable requirements concerning Static, Fatigue strength and Damage Tolerance will be met and the Work Package weight is optimized.

The process of structure substantiation includes:

- The writing of Sizing Sheets (SSs) for the Critical Elements (CrEIs) defined during the Design Phase, that will be improved and updated in Substantiation Forms (SFs) during Detailed Design Phase then gathered in Certification Documents (CDs) in all the substantiation domains as static, fatigue, damage tolerance, dynamic, flutter and emergency situations (as emergency landing, ditching, bird strike or engine burst). Certification documents will follow specified DASSAULT numbering and plan incorporating, in particular, a summary of sizing load cases, methods used, Safety Margins which will be provided in a dedicated Stress substantiation documentation, engineering procedure and Load Severity Indexes (LSI) and their allowable limits to assess the severity of a load case for a given structural zone without re-running the whole analysis.
- The definition and realization of partial structural tests deemed necessary when material or technology data are missing or when analysis requires an experimental support.
- The realizations of static, fatigue and damage tolerance (including residual strength) testing on a complete representative empennage structure (HTP & VTP).
- Operation tests according to §25.683 (b).
- The use of the first A/C to run structural tests (proof test, vibration tests, flight instrumentation calibration) before flight tests with the goal of collecting data to check and improve the load and structure database.
- Flight tests that allow demonstrating the flight envelope and gathering data for loads model actualization.
- A Waiver Acceptation Criteria Catalogue with corresponding structural substantiation for the manufacturing phase.
- Structural Substantiation required Data for validation of Structural Repair Manual (SRM), Modifications or Service Bulletins (SBs).

2.1 Tee Fittings Justification Strategy

The baseline definition of Tee fittings semi-product in the early stages of the program was die forging.

Due to problems in the supply chain, forged products will not be available for early aircraft. To address this problem, the following plan has been put in place:

- Tee fittings will be machined from plate from aircraft S/N 1 until S/N 5.

- From aircraft S/N 6 and up, Tee fittings will be forged and then machined to the final shape.
- Tee fittings to be installed in the HTP component test specimens (both static and fatigue) will be machined from plate.

The Type Certificate configuration shall consider the machined Tee fittings and a modification shall be done to change them to forging for S/N 6 aircraft and beyond. Following strategy is proposed to substantiate this modification:

<i>Domain</i>	<i>Reconciliation strategy proposal</i>
Static	1. Correlation of stress levels between test and FEM models used for analysis. 2. Update the analyses for the changeover to forging with the results of the correlation and the eventual change in material mechanical properties and allowable values.
Fatigue and damage tolerance	3. Correlation of stress levels between test and FEM models used for analysis. 4. Correlation of crack growth methodology by including an artificial damage in a critical area of the specimen and the analysis of its evolution. 5. Update the analyses for the changeover to forging with the results of the correlations.

Table 1. Tee fittings reconciliation strategy

3 Structural Stress Analysis (Internal Loads and FEM Model)

Structural analysis is performed by means of calculations from FEM internal loads, this FEM is the support tool of the data transfer between aircraft model and the work packages sub assembly model. During the JDP, AIRBUS D.S. has built up the Horizontal Stabilizer and Elevator F.E. general model (FEGM) (including mass distribution of the equipped ship set) in agreement with the boundary work packages. DASSAULT collected all the different WP FEGM delivered models, perform the assembly of the complete aircraft model and introduce additional mass (fuel, payload...), aerodynamic and systems data to compute more refined loop loads. A continuous process will be settled to improve aero servo elastic model (ASEM) all along the airplane conception from initial sizing studies to final certification process:

- After Feasibility Phase: DASSAULT has generated GFEM M2.00 with preliminary Loop load 0.0
- Sizing phase: AIRBUS D.S. has performed the HTP & Elevator preliminary Sizing with GFEM M2.0 – Loop 0.0, and is providing any WP GFEM significant change to DASSAULT that will integrate the changes into the new GFEM M2.10 – Loop 1.0 with available Wind Tunnel Test Results. AIRBUS D.S. will update the Sizing studies with this new loop loads and communicate to DASSAULT any WP GFEM significant change.
- Detailed Design Phase: DASSAULT will update the GFEM M2.11 with the different WP's changes and will generate the Loop load 1.1; then AIRBUS D.S. will perform the HTP & Elevator Detail Design with this GFEM M2.11 – Loop 1.1, and will write the Substantiation Forms for First Flight Clearance (5 months before the First Flight).
- Certification Phase: After First Flight Clearance, AIRBUS D.S. will provide any final WP GFEM significant change to DASSAULT in order to generate Aircraft GFEM M2.2 according to Drawing and Stress Substantiation documents, and with data from different Test, and After First Flight DASSAULT will issue the Certification load report with M2.2 – Loop 2.0.
 - If this Loop Loads are shown lower than the HTP and Elevator allowable limits demonstrated in the substantiation reports, there will not be new load delivery for HTP and Elevator and DASSAULT will update the Synthesis Substantiation document at aircraft level.
 - If this Loop Loads are shown greater than the HTP and Elevator allowable limits demonstrated in the substantiation reports, AIRBUS D.S. will rework the analysis, will perform dedicated test or Define and Substantiate a Design Change.

Static, fatigue and damage tolerance analyses may also be performed on this general model and complemented, if required, by refined FE analysis and tests at the coupon or components level.

The Horizontal Stabilizer and Elevator Global FEM will be produced using CATIA-ELFINI V8 or V9 code, and for detailed local models will be in 3DEXP format for windows; MSC/NASTRAN code will be also used for both Global and Detailed FEM at AIRBUS D.S. internal level.

The mesh size of the model is defined to represent adequately the stiffness of the different load paths, weight and inertia distribution (structural masses and others).

The nodes, elements and degrees of freedom of the models will be defined to avoid near singularity and spurious low frequency modes. For the free-free model, all six rigid body modes will have to be represented.

The model will include an adequate distribution of structural and non-structural mass.

The nodal points and elements will be defined and numbered to allow the integration into the overall aircraft GFEM. AIRBUS D.S. will be responsible for the Horizontal Stabilizer and Elevator GFEM. DASSAULT, or other supplier selected for this purpose, will integrate the overall model to obtain interface loads.

4 Material Allowable

AIRBUS D.S. has proposed a selection of materials based on AIRBUS certification, production and in-service experience (see Ref. 5).

Material data and static allowable for metallic materials are taken from the information contained in MIL-HDBK-5H or MMPDS-03.

Fatigue strength and fracture mechanics properties (S-N and da/dN curves and fracture toughness data) for metallic materials will be obtained from the AIRBUS D.S. database. Mean or average values will be used in accordance with AMC 25.571.

For metallic static analyses, B-values are used for multiple load path structures that meet the residual strength requirements with one load path failed. In other cases, A-values will be used.

For Composite Materials, the allowable is taken from AIRBUS D.S. database. B-basis and H/W (70° / 85% RH) data will be considered. With regards to the elastic moduli, mean values shall be used to perform stiffness and internal loads analyses and to convert internal loads into strains when needed for strength checks (e.g. in the FEM models) but for buckling analyses, B-basis values of the elastic modulus shall be used instead.

Material strength properties shall take into account any effect of protective treatment and all relevant manufacturing processes based upon reliable data.

Service temperature and environmental conditions are considered for material properties and F.E.M. analyses in accordance with airworthiness requirements and approved means of compliance.

The proposed material selection will be reviewed during the definition phase until CDR with DASSAULT, which could request some changes resulting from its own experience or for harmonization on the overall aircraft.

Fasteners and standard parts will be selected from DASSAULT standard database. Fastener selection shall minimize the number of types in order to minimize maintenance tooling and spare parts according to Ref. 3 requirements. AIRBUS D.S. can propose a specific fastener selection to DASSAULT during the CDR.

All materials and standard parts will be listed and compiled in a Material Engineering Report that will be referenced in the Detailed Technical Specification.

AIRBUS D.S. will propose a corrosion protection plan, according to Ref. 3 protection principles, for both metallic and composite parts to DASSAULT, which could request some changes, during the definition phase.

5 Static Strength

Static strength is proved by analysis, based upon finite element internal loads provided by DASSAULT from the Finite Element General Model (FEGM) of the complete aircraft, and failure criteria using reliable methods supported by tests (material characterization, coupon, sub-component, component and final airframe structural tests).

It is demonstrated that the structure withstands:

- Limit Loads (LL) without detrimental deformations
- Ultimate Loads (UL) without failure.

Where: UL are LL multiplied by appropriate safety factor (generally 1.5) and all special factors required by regulations (casting, fitting, others...) or other supplemental factor specified by DASSAULT.

Margins of safety or Reserve Factors shall be calculated in all structural elements by comparison of the Allowable failure criteria and the value of the Criteria at ultimate load:

$MS = 100 \times [(Allowable\ criteria / Criteria\ at\ ultimate\ load) - 1]$ or

$RF = [(Allowable\ criteria / Criteria\ at\ ultimate\ load)]$

Where the allowable failure criteria is obtained either:

- Reliable methods based on AIRBUS D.S. stress manual.
- Detail and subcomponent test already performed on similar certified structures.
- New detail and subcomponent test to be performed within Falcon 10X certification process.

All Margins of Safety shall be greater than zero or all Reserve Factors greater than one.

Following DASSAULT requirement, only Margins of Safety shall be provided in the synthesis report.

6 Sizing Criteria and Methods of Calculation

Structural elements will be sized and justified in accordance with the following criteria and methods of calculation (see Ref. 5).

	Item	Specific Sizing Criteria		Method of Calculation
METALLIC PARTS	Lugs	Strength	No yield under limit load Ultimate strength / bearing	OREJETAS
		Fatigue / DT	Stress below fatigue / Damage Tolerance allowable	#
	Fittings	Stability	No buckling under ultimate load No crippling under ultimate load	CRIPPLING #
		Strength	No yield under limit load Ultimate strength	FLEXA #
		Fatigue / DT	Stress below fatigue / Damage Tolerance allowable	#
	Joints	Strength	No yield under limit load Ultimate strength / bearing	ARES, BOLTS3D and BULONES #
		Fatigue / DT	Stress below fatigue / Damage Tolerance allowable	#

In addition to above referred methods, detailed FEM models and hand calculations based on well-known bibliography references as Bruhn, Roark, Niu, etc. will be also used.

Table 2. Sizing criteria and methods of calculation for metallic structure

	Item	Specific sizing criteria		Method of calculation
COMPOSITE PARTS	Main box skin covers	Stability	No global buckling under ultimate load No local buckling of stringer web under ultimate load No local buckling of skin sub-panels under limit load	ARPA
		Strength	Maximum fibre strain limited to damage tolerance allowable	
	Elevator skins	Stability	No buckling under ultimate load No sandwich typical local instabilities under ultimate load	SANDRES #
		Strength	Maximum fibre strain limited to damage tolerance allowable No sandwich typical local failures under ultimate load	
	Spars and ribs (main box and elevator)	Stability	No buckling under ultimate load No crippling (and inter-rivet buckling in riveted stiffeners) under ultimate load	PANPA and CRIPCO #
		Strength	Plain webs: maximum fibre strain limited to damage tolerance allowable Holed webs: maximum strain around the edge of the hole limited to maximum strain allowable	#
	Leading and trailing edges	Stability	No buckling under ultimate load (under limit load for the LE skin) No buckling of rib webs under ultimate load No crippling of rib flanges under ultimate load	PANPA and CRIPCO #
		Strength	Ultimate strength (tension/compression) of rib flanges and webs Ultimate strength of L/E nose against bird strike	#
	Joints	Strength	Ultimate bearing-bypass strength interaction	ARES and URCO

In addition to above referred methods, detailed FEM models and hand calculations based on well-known bibliography references as Bruhn, Roark, Niu, etc. will be also used.

Table 3. Sizing criteria and methods of calculation for composite structure

7 Fatigue Strength

Fatigue analysis will be performed for the most fatigue sensitive items in order to verify the Design Service Goal (DSG = 20000 flights or 60000 flight hours).

It will be demonstrated that the metallic structure of the Horizontal Stabilizer will have no crack initiation during the full aircraft Design Service Goal (DSG).

It will be also demonstrated a Limit of Validity (LOV) during which the metallic structure of the Horizontal Stabilizer will be free of Widespread Fatigue Damage (WFD). The first candidate LOV will be the DSG. A list of elements (if any) susceptible to WFD and their corresponding maintenance tasks (if any) will be included in the certification documentation as part of the WFD evaluation.

Fatigue load spectrum will be provided by DASSAULT as a combination of basic FEM loading according to DASSAULT Strength Substantiation Directives (Ref. 2), and will account for the loading conditions within the flight profile: taxi, take-off, climb, cruise, descent and ground.

The fatigue justification will be performed by analysis supported by test evidence. The calculation methods to be used will be the AIRBUS D.S. ones, described in Ref. 5. Applicable Scatter Factors are those provided by DASSAULT in Ref. 7.

Acoustic fatigue behaviour for metallic elements will be justified by analysis. For composite parts, the analysis could be supported by existing specific tests on similar structural design concepts. Acoustic loads will be provided by DASSAULT

The fatigue and damage tolerance of composite parts in the F10X will be justified by the non-initiation and non-growth concept. It will be substantiated through the empennage full-scale fatigue test that shall include artificial defects and damages.

During the definition phase this concept will be guaranteed by limiting the allowable static and fatigue strain levels to adequate maximums based on AIRBUS experience in testing coupons (*) and components of certified airframes.

(*) Including residual strength CAI tests but also tests with impacted specimens under cycling fatigue loads, showing that cycling strain levels around 50% of static strength shall no cause damage growth.

All these methods, software and processes follow the same DASSAULT approved principles described in Ref. 5.

8 Damage Tolerance Justification

The following damages will be considered for the damage tolerance justification:

- Fatigue.
- Environmental damage.
- Accidental.
- Manufacturing.
- Bird impact.
- Lightning.
- Windmilling after fan blade failure.
- Hail and other probable damages accounted for in the damage assessment threat concerning composite parts.

8.1 Metallic Structure

The structure will be designed in accordance with two damage tolerance objectives:

- Certification objective: No catastrophic failure shall result from any possible damage during the entire life of the aircraft.
- Economic objective: Maintenance tasks due to possible damages (inspections, repairs or replacements) will be as low as possible and in accordance with aircraft needs.

Crack growth analyses will be performed using the AIRBUS D.S. calculation methods together with the load spectra defined by DASSAULT.

Critical crack lengths will be calculated with the AIRBUS D.S. methods in order to obtain, together with crack growth curves, the thresholds and inspection intervals covering the objectives defined by DASSAULT in the Airframe General Technical Requirements (AGTR).

The crack growth and residual strength analyses will be supported by the data gathered during the Full Scale Fatigue and Damage Tolerance Test.

8.2 Composite Structure

Damage tolerance of the composite structure shall be substantiated by analysis, building block testing from the coupon to the full-scale levels and experience on similar structures according to Ref. 6. Following damages shall be accounted:

- Category 1: allowable damages that may go undetected by scheduled or directed field inspection and allowable manufacturing defects.

Ultimate Load capability and no-growth under repeated loads for a lifetime shall be demonstrated. Both demonstrations shall be supported by the full-scale test.

Damages to be considered in this category shall be:

- Allowable manufacturing flaws such as delamination, porosity and defect in the bond lines.
- Allowable scratches/nicks.
- Undetectable damages due to accidental impacts (BVID) up to an energy of 35 joules.

- Category 2: damages that can be reliably detected by scheduled or directed field inspections performed at specified intervals.

Residual strength capability at Limit Load and no-growth under repeated loads in, at least, two inspection intervals shall be demonstrated. Both demonstrations shall be supported by the full-scale test.

Damages to be considered in this category shall be:

- Visible damages due to accidental impacts (VID) up to 80 joules.

- Category 3: damages that can be reliably detected within a few flights of occurrence by operations or ramp maintenance personnel without special skills in composite inspection.

Residual strength capability at Limit Load shall be demonstrated supported by the full-scale test.

Damages to be considered in this category shall be:

- Large visible impact damages (LVID), typically penetrating holes.
- Significant lightning strike damages (*).

(*) Although severe lightning damages are category 4, residual strength capability at Limit Load shall be demonstrated by analysis supported by the lightning tests and the empennage full scale structural test.

Additionally, large damage and fail-safe capabilities shall be addressed by considering following damages:

- Large cut-outs/holes (e.g. severed stringer and/or adjacent skin bay, large hole in compression panel or severed spar chord).
- Large disbonding (e.g. complete sandwich face sheet disbonding between arrestment features). (TBC)
- Removal of fasteners from critical joints/fittings.
- Disconnection of one hinge in multi-hinged movables.

All this type of damages shall be introduced in the full-scale test specimen prior to the residual strength phase for the composite structure (see section 9.2.3.3) except:

- Large cuts/holes in the skin covers which shall be substantiated at a lower pyramid level through the panel testing (see section 9.1.2)
- Large cut in the spar chord which shall be substantiated by analysis (**) supported by AIRBUS D.S. experience in similar structures (see Ref. 9). The allowable value shall be derived from testing at coupon level (see section 9.1.1).

(**) Notch fracture analysis plus F.E. load redistribution analysis.

- Category 4: discrete source damages obvious to the flight crew (allowing flight manoeuvres to be limited until landing) requiring immediate repair after flight.

Residual strength capability at “get home loads” for the duration of the flight shall be demonstrated by analysis.

Damages to be considered in this category shall be:

- Bird strike.
- Severe in-flight hail.

8.2.1 Detectability Criteria

According to AIRBUS D.S. policies, the metric to define the impact damage thresholds shall be the following:

<i>Type of damage</i>	<i>Type of inspection</i>	<i>Metric</i>	<i>Detectability thresholds</i>
BVID	General visual inspection (GVI)	Dent depth	1.3 mm (*)
		Crack length (single or group)	70 mm
LVID	Walk around	Penetrating hole	
(*) After relaxation. According to AIRBUS D.S. experience, 1.3 mm dents following the impact would correspond to dents depth after impact between 2 and 2.5 mm for carbon fibre with epoxy resin laminates.			

Table 4. Criteria of detectability for BVID and LVID

9 Structural Tests

9.1 Development Tests

Development tests are focused on risk minimization. Their main purpose is to validate the applicability of AIRBUS D.S. analytical methods and failure criteria to the specific characteristics of the Horizontal Stabilizer structure design.

The following development test will be carried out.

9.1.1 Coupons with Centred Transverse Crack

Coupons with centred full-penetrating transverse cracks shall be tested according to following test matrix in order to obtain the corresponding residual strength allowable. This allowable (average value under RT/AR conditions) shall be used in the analysis justification of the main box spar chords residual strength following a category 3 large cut.

<i>Property</i>	<i>Material</i>	<i>Laminates</i>	<i>Damage</i>	<i>Test Temperature</i>	<i>Ageing</i>	<i>Number of specimens</i>
Plain tension strength and modulus	UD-Tape Hexcel IMA/M21E (grade B)	L1	None	RT	AR	3
Residual tension strength			a1			6
			a2			6
			a3			6

where “ai” stands for different crack lengths.

9.1.2 Skin Panels

Two panels representative of the actuator access cut-out area in the torsion box lower skin will be tested in RT/AR conditions. One of the panels will be tested in pure compression and the other one in pure shear. Both of them shall include category 1 damages (BVID limited to an energy of 35 joules).

No generic stiffened panels shall be tested to support the substantiation of the HTP skins typical design. It shall be covered by the VTP panel testing as both main box skins are completely similar (i.e made of the same material and processes (cocuring) and designed following the same requirements and principles). The object of these panel tests is the validation of the methodology and the small differences can be covered with the VTP development tests (see Ref. 8).

9.1.3 HTP-VTP Fairing Functional Test

One specimen representative of the spring loaded fairing design shall be tested with following objectives:

- Validation of the fairing concept.

- Demonstration of compliance with the design requirements (i.e. shape, stiffness and strength at Limit Load).

9.1.4 Lightning Strike Tests

Lightning development test will be performed in several configurations of solid and sandwich CFRP in order to evaluate the behaviour of different lightning protections and different paint scheme of 240 µm and 400 µm.

These configurations shall be subjected to lightning tests at zones 1A, 2A (severe and frequent strike), 1B & 2B (severe strike) according to applicable standards ARP 5416/ED-105.

9.1.5 Test on HTP-VTP Y-link

Due to the complexity of this joint, static and fatigue tests shall be performed to validate the design concept.

Test details are in progress and shall be included in further issues of this report.

9.2 Certification Tests

9.2.1 Lightning Strike Tests

Lightning certification tests will be performed to show compliance with the certification requirements given in §25.581 and §25.899.

Three (3) test specimens representing different cut outs of the HTP Torsion Box Upper Skin shall be tested.

Specimen TB-1 representing the outer extreme of the HTP Torsion Box, with 12 plies. Then, the obtained results in terms of physical damages can be used to evaluate the 12 CFRP plies areas as well as other thicker areas of the HTP Torsion Box where resulting lightning damage would be consequently smaller.

Specimen TB-2 representing a local area of minimum skin thickness of the HTP, 6 CFRP plies. Then, the results will be used to justify the small areas with a minimum skin thickness but less structurally demanded.

Specimen TB-3 representing the inner part of the HTP Torsion Box, 26 plies. The initial conservative approach is to use the damages of specimen TB-1 for justifying the structural integrity of all the areas of the HTP/VTP torsion box equal or thicker than 12 CFRP plies. However, specimen TB-3 is manufactured and tested in order to have damage data of thicker skins for justifying the structural integrity in case of specific locations where applying resulting damage from specimen TB-1 is too conservative.

The construction of the HTP Torsion Box is similar than the VTP Torsion Box, therefore, test results can be used for structural analysis purposes of both components. Additionally, the VTP skin thickness are equal/higher than HTP skin thickness.

The specimens shall include:

- Three integrated internal co-cured T-shape stringers. The distance between stringers similar to the actual design.
- The external LS protection. One ply of 73 gr/m² ECF co-cured at outer face of the CFRP skin.

Additionally, the TB-1 shall include bolted aluminium plate representing the joint to the actual HTP main box end rib.

The test specimens shall be painted with the same painting scheme than the series aircraft. The final paint thickness shall be 220±20µm to cover nominal paint scheme expected after complete manufacturing process.

The test specimens shall be tested against zone1A and zone2A level in accordance with the lightning zoning (defined according to ED-91). The simulated lightning test waveforms to be applied shall be in accordance with ED-84. The test shall be conducted following the guidance provided by ED-105.

See Ref. 11 for further details.

9.2.2 Leading Edge Bird Impact Test

A certification bird impact test of the inboard leading edge (the outboard leading edge will be justified by similarity as it is less critical in terms of geometry) will be carried out in the following conditions:

- The specimen shall include a representative attachment of the Leading edge to the CFRP Lateral Box including the front spar.
- The test jig shall be a heavy rigid structure.
- Bird impact of 8 pounds at Vc (0) = 300 knots according to §26.631.
- Test shall be conducted at laboratory conditions with the specimen as received (RT/AR).

9.2.3 Empennage Full Scale Test

The complete empennage (VTP & HTP) shall be tested in the same article.

The F10X empennage is a hybrid structure (roughly speaking, carbon fibre torsion boxes and metallic fittings). For this kind of structures, AIRBUS D.S. philosophy is to perform the static strength, fatigue and damage tolerance justifications in the same test. However, given the complexity of replacing some of the Horizontal Stabilizer metal parts (e.g. rear fitting and centre splices), two specimens of this particular structure shall be finally used.

Due to the dissimilar requirements of composite and metallic structures regarding repeated loads demonstration, the following general sequence shall be followed

- Fatigue test of the composite structure.
- Static test of the metallic and composite structure.
- Damage tolerance and residual strength test of the composite structure.
- Replacement of following elements in the test article:
 - The complete Horizontal Stabilizer.
 - Those fatigue-sensitive metallic parts of the Vertical Stabilizer structure.
- Fatigue test of the metallic structure.
- Damage tolerance and residual strength test of the metallic structure.

Then, regarding the Horizontal Stabilizer structure, two different specimens will be used to justify the static strength, fatigue and damage tolerance whereas, with regards to the Vertical Stabilizer, only one specimen of the composite structure and two of the metallic will be tested.

The fatigue loads for the phases to justify the composite and the metallic structure, will be obtained with the different appropriate criteria:

- Composite: No clipping of high spectra peaks. Omission of cycles below 30% of limit load.
- Metal: Clipping of high spectra peaks to 70-80% of limit load. Omission of cycles below a certain value to assure a reasonable test duration. Damage produced by omitted cycles compensated by increasing level of cycles not omitted.

9.2.3.1 Test Specimen Description

The test article shall include:

- Main torsion boxes (HTP (both sides) & VTP).
- Leading edges (HTP (both sides) & VTP).
- Elevator, both sides; rudders.
- Fittings and relevant system supports.

The specimen shall incorporate:

- Artificial defects representative of the allowable manufacturing flaws in composite structure.
- Impact damages in the composite structure until BVID (with energies not higher than 35J) introduced at the very beginning of the test, VID (with energies not higher than 80J) introduced before the damage tolerance phase and large damages introduced before the residual strength checks.
- In-shop and in-service repairs (TBC the latter).

These defects, damages and repairs will be introduced in the most critical areas.

9.2.3.2 Test Conditions

Test shall be conducted at laboratory conditions and with the specimen as received (RT/AR).

According to the state of the art in certification processes of composite structures, the environmental effects shall be covered at lower levels of the test pyramid and therefore no E-LEF shall be applied to the static loads in this test.

In this respect, this full-scale test shall be used only to demonstrate that the strains at critical locations caused by the application of limit and ultimate load match the predicted strains and are no greater than the applicable allowable or design values for the critical environment.

Regarding fatigue, AIRBUS has shown (during the development and certification of the A350) that, for IMA / M21E material, aging does not affect the repeated loads justification.

9.2.3.3 Test Sequence

The test shall be carried out in the following sequence:

- Introduction of category 1 damages in composite parts (see section 8.2).
- Tests at 70% Limit Load for first flight clearance:
 - Functional test to show compliance with §25.683.
 - Static test (number of test cases to be agreed with DASSAULT).
- Functional tests at Limit Load to show compliance with §25.683.
- Static Limit Load test of the complete set of critical cases.
- Fatigue phase for composite structure: fatigue loads corresponding to 2/3 life (13300 flights) with a LEF of 1.17 (*).
- Static Ultimate Load test.
- Introduction of category 2 damages in composite parts (see section 8.2) (**).
- Damage Tolerance phase for composite structure: fatigue loads corresponding to 1/3 life (6700 flights) with a LEF of 1.17 (*).
- Introduction of category 3 damages (see section 8.2).
- Residual strength test for composite parts at Limit Load.
- Repair of composite structure, if necessary, to use it as a dummy in the fatigue test of metallic parts.
- Replacement of following metallic parts in the test specimen:
 - The complete Horizontal Stabilizer.
 - Those fatigue-sensitive metallic parts of the Vertical Stabilizer structure.
- Fatigue phase for metallic structure: 2 lives (40000 flights) minimum.
- Introduction of artificial damages in metallic parts including disconnections.
- Damage Tolerance phase: crack growth and residual strength test for metallic parts.
 - Fatigue cycling corresponding to $3 \times II$ (4000 flights) = 12000 flights.
 - Limit load application with artificial damages.
 - Reconnection of disconnected load paths and repair of metallic parts as considered necessary.
- Decision about test continuation: static margin search test and/or failure test of the complete structure.
- Tear down inspection (TBC).

(*) A load enhancement factor (LEF) of 1.17 to cover composite material scatter shall be applied to the fatigue and damage tolerance loads.

(**) As a “good engineering practice”, those category 3 damages in composite parts aimed at demonstrating fail-safe capability (see section 8.2) and not easy to detect from the outside may be introduced at this stage.

9.2.3.4 Test Supporting Structure and Loading System

Since the interface reactions of the complete empennage shall be validated in the full-scale test of the complete aircraft structure (A/C MTC), the test article may be statically determinate supported to simplify and reduce the cost of the rig. Missing reactions shall be introduced by means of actuators. In this respect, this test shall be used just to validate the FEM model.

The loading system shall be designed so as to assure that the overall internal load distribution and interface reactions are realistic and slightly conservative.

10Planning

The following tables summarise the planned dates for the main milestones of the programme (including tests). All dates are in accordance with Ref. 10 except for the FFRR which was updated during the FST workshop held in Paris on June 8th 2022.

MILESTONE	COMMENTS	DATE
PDR		24/11/2020
PGR		Mid Feb 2021
CDR		July 2022
CGR		October 2022
FF	TO BE CONFIRMED BY DA	Jan-June 2024
CERTIFICATION	TO BE CONFIRMED BY DA	Mid 2025
LOOP 0 FOR PDR		sep-19
LOOP 1.0 FOR CDR		12/01/2021
LOOP 1.1 FOR FST		Feb-June 2022
LOOP 2 FOR CERTIFICATION	TO BE DEFINED BY DA	8 months after FFRR
PDR CrEIs SIZING	Loop 0	01/06/2020
PDR CrEIs SIZING (data needed for Flutter)	Loop 0	20/04/2020
PDR CrEIs Justification Report	Loop 0	1 week before PDR
CDR SIZING	Loop 1.0 + Loop 1.1(some elements)	June 2022
CDR Justification Report	Loop 1.0	June 2022 / January 2023
Loop 1.0 / Loop 1.1 Exceedance check		Beginning 2023
FST load Check		TBC
FF Justification	Loop 1.1	1 month before FF
Loop 2.0 exceedances check	Pending on Loop 2.0 delivery	2 months after Loop 2.0 delivery
Certification Justification Report	Pending on Loop 2.0 loads delivery	2025 (TBC)

<i>HTP</i>	<i>TEST</i>	<i>SPECIFICATION</i>	<i>TEST DATE</i>
Development Tests	SHEAR PANEL (access hole)	TEA-F9-SP-200006-C	July 2022
	COMPRESSION PANEL (access hole)		October 2022
	HTP-VTP FAIRING FUNCTIONAL TEST	TEA-F9-SP-210003-A	First iteration: Jan. 2022 Second iteration: Jan. 2023
	TEST on HTP-VTP Y-LINK	TBD	Q3 2023
Certification Tests	BIRD IMPACT LE	TE-F9-SP-220001-A	Q3 2023
	FULL SCALE TEST	TE-F9-SP-220003-A	August 2024 (F.F. Clearance)

Table 5. Planning of main program milestones